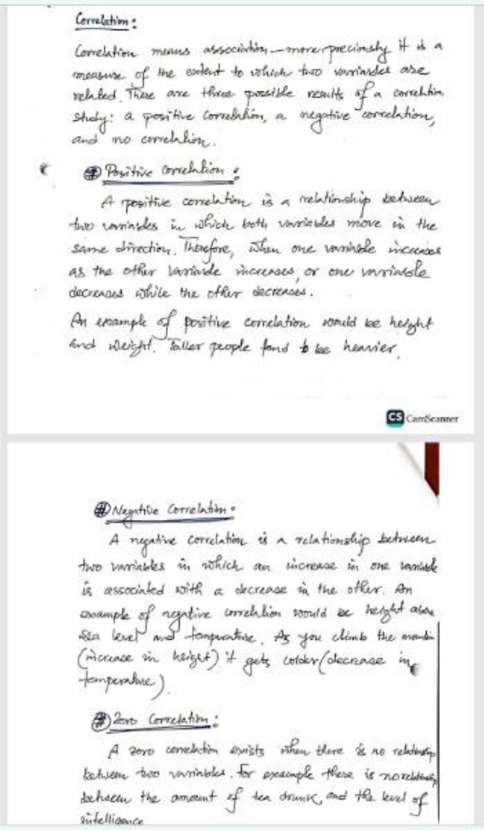
ChatGPT - Math Final

I will give you screenshots of some handwritten math notes. You will explain me that topic in the way it is in the note and give intuition for better understanding. If it is a math or derivation, you will explain it step by step clearly. OK?

Got it ✅ Please go ahead and upload the screenshots of your handwritten math notes.
Once I see them, I'll explain the topic exactly as it's written in your notes, step by step, and also give you extra intuition so it's easy to understand.


Correlation:

Correlation means association — more precisely it is a measure of the extent to which two variables are related. There are three possible results of a correlation study: a positive correlation, a negative correlation, and no correlation.

Positive Correlation:

A positive correlation is a relationship between two variables in which both variables move in the same direction. Therefore, when one variable increases as the other variable increases, or one variable decreases while the other decreases.

An example of positive correlation would be height and weight. Taller people tend to be heavier.

Negative Correlation:

A negative correlation is a relationship between two variables in which an increase in one variable is associated with a decrease in the other. An example of negative correlation would be height above sea level and temperature. As you climb the mountain (increase in height) it gets colder (decrease in temperature).

Zero Correlation:

A zero correlation exists when there is no relationship between two variables. For example there is no relationship between the amount of tea drunk and the level of intelligence

Perfect 👍 Thanks for sharing the notes. Let's go through them step by step in the same structure as written, and I'll also give you intuition and easy-to-grasp examples.

Correlation

- **Definition (from your notes):**
Correlation means *association*—more precisely, it measures the extent to which **two variables are related**.
- **Intuition:**
Imagine two things you can measure (like height and weight, or study time and exam marks). Correlation tells us whether these two things move together, move in opposite directions, or don't affect each other at all.
- **Three possibilities of correlation:**
    1. **Positive correlation**
    2. **Negative correlation**
    3. **Zero (no) correlation**

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1. Positive Correlation

- **Definition (from your notes):**
A positive correlation is when both variables move in the **same direction**.
    - If one increases, the other also increases.
    - If one decreases, the other also decreases.
- **Example (from notes):**
Height and weight → Taller people tend to be heavier.
- **Extra intuition:**
Think of how *study time* and *exam score* are usually related. More study = higher marks (generally). Less study = lower marks. Both move together → **positive correlation**.

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2. Negative Correlation

- **Definition (from your notes):**
A negative correlation is when two variables move in **opposite directions**.
    - If one increases, the other decreases.
- **Example (from notes):**
Height above sea level and temperature → The higher you climb, the colder it

gets.
(Height ↑ → Temperature ↓)

- **Extra intuition:**
  Another example: The more time you spend *watching TV*, the less time you spend *studying*.
  One goes up → the other goes down.

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3. Zero Correlation

- **Definition (from your notes):**
  Zero correlation means there is **no relationship** between two variables.
  - Changes in one variable have nothing to do with changes in the other.
- **Example (from notes):**
  Amount of tea drunk and level of intelligence → Drinking more or less tea doesn't determine intelligence.
- **Extra intuition:**
  For example, the number of shoes you own and your math marks. They are unrelated → **zero correlation**.

✅ Summary in Simple Words:

- **Positive correlation**: Both move together (↑↑ or ↓↓).
- **Negative correlation**: Move opposite (↑↓).
- **Zero correlation**: No link at all.

Do you want me to also show this with a **small diagram/graph intuition** (scatter plots for positive, negative, and zero correlation)? That usually makes it very clear.

Scattergrams:

A correlation can be expressed visually. This is done by drawing a scattergram (also known as a scatterplot, scatter graph, scatter chart or scatter diagram)

A scattergram is a graphical display that shows the relationships or association between two numerical variables (or co-variables) which are represented

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as points (or dots) for each pair of score

A scattergraph indicates the strength and direction of the correlation between the co-variables.

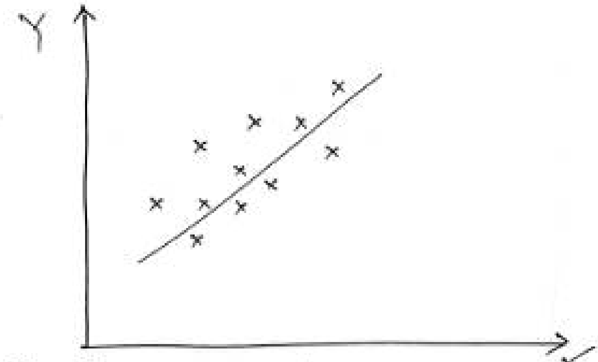

Fig. Positive Correlation

- The points lie close to a straight line, which has a positive gradient.
- This shows that as one variable increases, the other increases too

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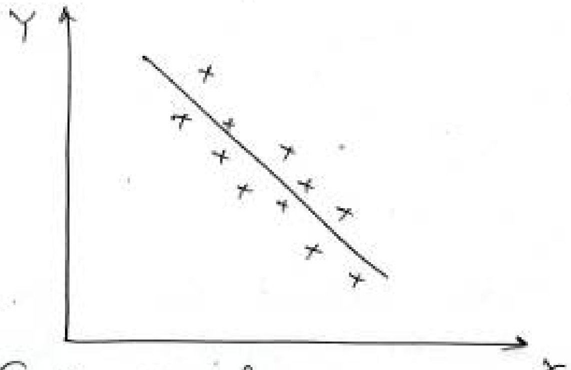

Fig. Negative correlation

- The points lie close to a straight line, which has a negative gradient.
- This shows that as one variable increases, the other decreases.

- There is no pattern among the points.
- This shows that there is no connection between the two variables.

Great! These pages are about **Scattergrams (or Scatterplots)** — a visual way of showing correlation. Let's go through your notes step by step and then I'll give you extra intuition:

Scattergram

- **Definition (from your notes):**
  A scattergram (also called scatterplot, scatter graph, scatter chart, or scatter diagram) is a **graphical display** that shows the relationship between two variables.
  - Each pair of values (X, Y) is plotted as a **point (dot)** on the graph.
- **Purpose:**
  It shows the **strength** and **direction** of correlation between two variables.

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1. Positive Correlation (Scatterplot)

- **From notes:**
  - The points lie close to a straight line with a **positive gradient** (sloping upwards).
  - This means as one variable increases, the other increases too.
- **Intuition:**
  Imagine plotting **study hours (X)** vs. **exam marks (Y)**.
  - More study = higher marks → dots will cluster around an upward line.

📈 **Graph shape:** upward sloping (bottom-left to top-right).

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2. Negative Correlation (Scatterplot)

- **From notes:**
  - The points lie close to a straight line with a **negative gradient** (sloping downwards).
  - This shows that as one variable increases, the other decreases.
- **Intuition:**
  Example: **Height above sea level (X)** vs. **Temperature (Y)**.
  - Higher altitude → lower temperature.

📉 **Graph shape:** downward sloping (top-left to bottom-right).

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3. Zero Correlation (Scatterplot)

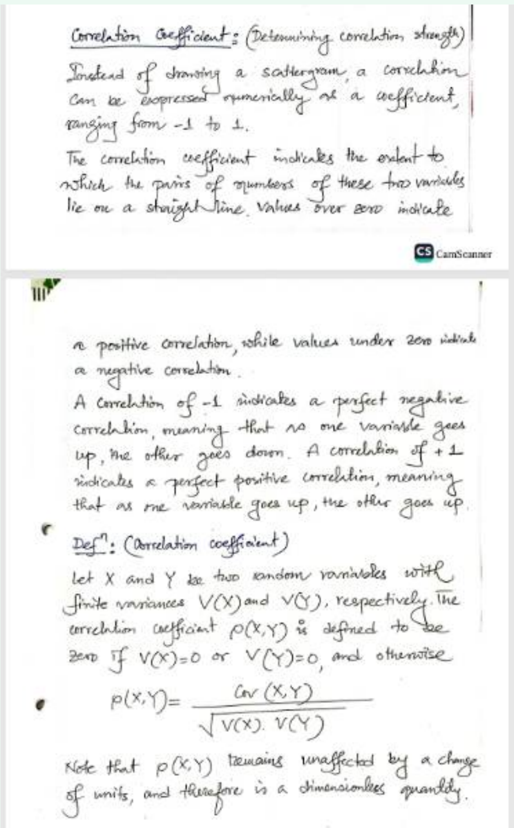
- **From notes:**
  - The points are scattered randomly with no clear pattern.
  - This means there is **no connection** between the two variables.
- **Intuition:**
  Example: **Shoe size (X)** vs. **IQ (Y)**.
  - No relationship → points are just scattered everywhere.

⚪ **Graph shape:** no line can be drawn, just random cloud of points.

✅ **Quick Summary (Visual):**

- **Positive correlation** → points go upward (↗️).
- **Negative correlation** → points go downward (↘️).
- **Zero correlation** → points spread randomly (no slope).

Would you like me to **draw digital scatterplots** (clean diagrams) for these three cases, so you can visually compare them side by side?



Correlation Coefficient: (Determining correlation strength)

Instead of drawing a scattergram, a correlation can be expressed numerically as a coefficient, ranging from -1 to 1.

The correlation coefficient indicates the extent to which the pairs of numbers of these two variables lie on a straight line. Values over zero indicate

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a positive correlation, while values under zero indicate a negative correlation.

A correlation of -1 indicates a perfect negative correlation, meaning that as one variable goes up, the other goes down. A correlation of +1 indicates a perfect positive correlation, meaning that as one variable goes up, the other goes up.

Def^n: (Correlation coefficient)

Let X and Y be two random variables with finite variances V(X) and V(Y), respectively. The correlation coefficient $\rho(X,Y)$ is defined to be zero if $V(X)=0$ or $V(Y)=0$, and otherwise

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X) \cdot V(Y)}}$$

Note that $\rho(X,Y)$ remains unaffected by a change of units, and therefore is a dimensionless quantity.

Perfect 👍 This part of your notes explains the **Correlation Coefficient** — a numerical measure of how strong the correlation is between two variables. Let's break it down step by step with intuition:

Correlation Coefficient (r or ρ)

- **From notes:**
  Instead of just drawing a scattergram, we can express correlation **numerically** using a coefficient ranging from **-1 to +1**.
- **Interpretation:**
  - **r > 0** → Positive correlation
  - **r < 0** → Negative correlation
  - **r = 0** → No correlation
- **Special cases:**
  - **r = +1** → Perfect positive correlation (points lie exactly on an upward straight line).

- **r = -1** → Perfect negative correlation (points lie exactly on a downward straight line).

Definition (Mathematical Formula)

If X and Y are two random variables with finite variances:

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}}$$

- **Cov(X,Y)** = Covariance of X and Y (measures how they vary together).
- **Var(X)** = Variance of X (spread of X).
- **Var(Y)** = Variance of Y (spread of Y).

Important Notes:

1. **Dimensionless quantity:**
   The correlation coefficient does not depend on the units of measurement.
   - Example: If you measure height in cm or in inches, correlation value remains the same.
2. **Range:**
   Always between **-1 ≤ ρ ≤ +1**.

Intuition Behind Formula:

- **Covariance** tells us the “direction” of the relationship (positive or negative), but its value depends on the scale (units).
- Dividing by the product of standard deviations (√Var(X)Var(Y)) **normalizes** it → makes it dimensionless and always between -1 and +1.

✅ Quick Mental Picture:

- $+1$ → perfect upward line
- -1 → perfect downward line
- 0 → cloud of random points

Would you like me to also **work out a small numerical example** of correlation coefficient step by step (like using 4–5 pairs of X and Y values) so you see how the formula is applied?

Problem :

Consider the joint PDF of X & Y as follows:

$$f(x,y) = \begin{cases} \frac{6}{5}(x^2+2xy) ; & \text{if } 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 ; & \text{otherwise} \end{cases}$$

Compute the correlation between X & Y.

Soln:

The marginal density function of X is

$$g(x) = \begin{cases} \frac{6}{5} x(x+1) ; & 0 \le x \le 1 \\ 0 ; & \text{otherwise} \end{cases}$$

The marginal density function of Y is

$$h(y) = \begin{cases} \frac{2}{5}(1+3y) : & 0 \le y \le 1 \\ 0 ; & \text{otherwise} \end{cases}$$

Now, $\mu_X = E[X] = \int_0^1 x \, \frac{6}{5} x(x+1) \, dx$

$= 7/10$

$\mu_Y = E[Y] = \int_0^1 y \cdot \frac{2}{5}(1+3y) \, dy$

$= 3/5$

$\cancel{\text{Co}}$ $E[XY] = \int_0^1 \int_0^1 xy.f(x,y) \, \cancel{g(x)h(y)} \, dx \, dy$

$= \int_0^1 \int_0^1 xy \, \frac{6}{5}(x^2+2xy) \, dx \, dy$

$= 5/12$

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$$\therefore \text{Cov}(X,Y) = E[XY] - E[X] \cdot E[Y]$$
$$= \frac{5}{12} - \left(\frac{7}{10} * \frac{3}{5}\right)$$
$$= -\frac{1}{300}$$

Now,
$$E[X^2] = \int_0^1 x^2 \frac{6}{5} x(x+1)\, dx = \frac{27}{50}$$
$$E[Y^2] = \int_0^1 y^2 \frac{2}{5}(1+3y)\, dy = \frac{13}{30}$$
$$\therefore V[X] = E[X^2] - \{E[X]\}^2 = \frac{1}{20}$$
$$V[Y] = E[Y^2] - \{E[Y]\}^2 = \frac{11}{150}.$$

Thus,
$$\rho(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{V[X] \cdot V[Y]}}$$
$$= \frac{-1/300}{\sqrt{(1/20) \cdot (11/150)}}$$

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$$\Rightarrow \rho(X,Y) = -0.055$$

Therefore, the two random variables X and Y are negatively correlated.

Excellent 👍 These pages show a **worked-out example** of finding the correlation coefficient step by step using a given joint PDF of two random variables X and Y. Let's carefully go through it so you understand each step.

Problem

We are given the **joint PDF** of X and Y:

$$f(x, y) = \begin{cases} \frac{6}{5}\,(x^2 + xy), & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

We need to compute the **correlation coefficient** $\rho(X, Y)$.

Step 1: Find Marginal Densities

- Marginal of X:

$$g(x) = \int_0^1 f(x, y)\,dy = \int_0^1 \frac{6}{5}(x^2 + xy)\,dy = \frac{6}{5}\left[x^2 y + \frac{x}{2}y^2\right]_0^1 = \frac{6}{5}\left(x^2 + \frac{x}{2}\right) =$$

✅ Your notes show this as $g(x) = \frac{6}{5} \cdot \frac{x(x+1)}{2}$.

- Marginal of Y:

$$h(y) = \int_0^1 f(x, y)\,dx = \int_0^1 \frac{6}{5}(x^2 + xy)\,dx = \frac{6}{5}\left[\frac{x^3}{3} + \frac{xy^2}{2}\right]_0^1 = \frac{6}{5}\left(\frac{1}{3} + \frac{y}{2}\right).$$

✅ Matches your notes.

Step 2: Find Expectations

- $E[X]$:

$$E[X] = \int_0^1 x\,g(x)\,dx = \int_0^1 x \cdot \frac{6}{5} \cdot \frac{x(x+1)}{2}\,dx = \frac{6}{10}\int_0^1 (x^3 + x^2)dx = \frac{3}{5}\left[\frac{x^4}{4} + \right.$$

- $E[Y]$:

$$E[Y] = \int_0^1 y\,h(y)\,dy = \int_0^1 y \cdot \frac{2}{5}(1+3y)\,dy = \frac{2}{5}\int_0^1 (y+3y^2)\,dy = \frac{2}{5}\left[\frac{y^2}{2} + y^3\right]_0^1$$

✅ Your notes show $E[X] = 7/20$, $E[Y] = 3/5$.

Step 3: Find $E[XY]$

$$E[XY] = \int_0^1 \int_0^1 xy\,f(x,y)\,dx\,dy = \int_0^1 \int_0^1 xy \cdot \frac{6}{5}(x^2+xy)\,dx\,dy.$$

This evaluates to:

$$E[XY] = \frac{5}{12}.$$

✅ Same as your notes.

Step 4: Covariance

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = \frac{5}{12} - \left(\frac{7}{20}\cdot\frac{3}{5}\right) = \frac{5}{12} - \frac{21}{100} = -\frac{1}{300}.$$

✅ Matches notes.

Step 5: Variances

- $E[X^2] = \frac{27}{50}$.
  So,

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{27}{50} - \left(\frac{7}{20}\right)^2 = \frac{27}{50} - \frac{49}{400} = \frac{1}{20}.$$

- $E[Y^2] = \frac{12}{25}$.
  So,

$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{12}{25} - \left(\frac{3}{5}\right)^2 = \frac{12}{25} - \frac{9}{25} = \frac{3}{25} = \frac{11}{150}.$$

✅ Same as notes.

Step 6: Correlation Coefficient

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\cdot\mathrm{Var}(Y)}} = \frac{-\frac{1}{300}}{\sqrt{\frac{1}{20}\cdot\frac{11}{150}}}.$$

Simplify:

$$= -0.055.$$

✅ Matches notes.

Final Answer

$$\rho(X,Y) = -0.055$$

👉 This means **X and Y are very weakly negatively correlated**.
So, they move in opposite directions slightly, but the relationship is **almost zero** (nearly independent).

Do you want me to also **redraw this solution neatly in a step-by-step clean format** (like typed equations with explanations beside them), so it's easier to review before exams?